

PART-A

Name of the Student: Bhupathi Nikhil Agnihotri

Signature of the Student: B. Nikhil Roll No.: 230123074

Signature of the Invigilator: [Signature]

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4	7	16

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$ and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Regression Assumptions allow the following

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{n \times 1} \quad X = \begin{bmatrix} x_1^1 & x_1^2 \\ x_2^1 & x_2^2 \\ x_3^1 & x_3^2 \end{bmatrix}_{n \times p} \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}_{p \times 1} \quad \left\{ \begin{array}{l} Y = X\beta \end{array} \right.$$

$$\min_{\beta} (Y - X\beta)^T (Y - X\beta)$$

$$\frac{\partial}{\partial x} (x^T A x) = (A^T + A)x$$

$$\frac{\partial}{\partial x} (a^T x) = a$$

$$\Rightarrow (Y^T - \beta^T X^T) (Y - X\beta)$$

$$\Rightarrow Y^T Y - \beta^T X^T Y - Y^T X \beta + \beta^T X^T X \beta$$

setting derivative
wrt $\beta = 0$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{or} \quad Y = X (X^T X)^{-1} X^T Y$$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 1 & 2 \\ -1 & -2 & -1 \end{bmatrix} \quad \left\{ \begin{array}{l} X^T X = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix} \end{array} \right.$$

$$(X^T X)^{-1} = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 2/9 \end{bmatrix} ; \quad X^T Y = \begin{bmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} 1/3 y_1 + 1/3 y_3 \\ 1/3 y_1 - 1/3 y_2 \end{bmatrix}$$

are the least square
estimates of θ_1, θ_2
— or — β_1, β_2 .

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Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$. Compute $A+B$ and $A-B$.

$$A+B = \begin{bmatrix} 1+9 & 2+8 & 3+7 \\ 4+6 & 5+5 & 6+4 \\ 7+3 & 8+2 & 9+1 \end{bmatrix} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 1-9 & 2-8 & 3-7 \\ 4-6 & 5-5 & 6-4 \\ 7-3 & 8-2 & 9-1 \end{bmatrix} = \begin{bmatrix} -8 & -6 & -4 \\ -2 & 0 & 2 \\ 4 & 6 & 8 \end{bmatrix}$$

$$A \cdot B = \begin{bmatrix} 1 \cdot 9 + 2 \cdot 6 + 3 \cdot 3 & 1 \cdot 8 + 2 \cdot 5 + 3 \cdot 2 & 1 \cdot 7 + 2 \cdot 4 + 3 \cdot 1 \\ 4 \cdot 9 + 5 \cdot 6 + 6 \cdot 3 & 4 \cdot 8 + 5 \cdot 5 + 6 \cdot 2 & 4 \cdot 7 + 5 \cdot 4 + 6 \cdot 1 \\ 7 \cdot 9 + 8 \cdot 6 + 9 \cdot 3 & 7 \cdot 8 + 8 \cdot 5 + 9 \cdot 2 & 7 \cdot 7 + 8 \cdot 4 + 9 \cdot 1 \end{bmatrix} = \begin{bmatrix} 30 & 28 & 22 \\ 78 & 70 & 64 \\ 114 & 106 & 100 \end{bmatrix}$$

$$B \cdot A = \begin{bmatrix} 9 \cdot 1 + 8 \cdot 4 + 7 \cdot 7 & 9 \cdot 2 + 8 \cdot 5 + 7 \cdot 8 & 9 \cdot 3 + 8 \cdot 6 + 7 \cdot 9 \\ 6 \cdot 1 + 5 \cdot 4 + 4 \cdot 7 & 6 \cdot 2 + 5 \cdot 5 + 4 \cdot 8 & 6 \cdot 3 + 5 \cdot 6 + 4 \cdot 9 \\ 3 \cdot 1 + 2 \cdot 4 + 1 \cdot 7 & 3 \cdot 2 + 2 \cdot 5 + 1 \cdot 8 & 3 \cdot 3 + 2 \cdot 6 + 1 \cdot 9 \end{bmatrix} = \begin{bmatrix} 88 & 89 & 90 \\ 46 & 53 & 56 \\ 14 & 19 & 24 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}, B^T = \begin{bmatrix} 9 & 6 & 3 \\ 8 & 5 & 2 \\ 7 & 4 & 1 \end{bmatrix}$$

$$\begin{aligned} & \begin{cases} 0 = (1, 0, 0) \cdot \vec{v} \\ (1, 0, 0) \cdot \vec{v} = (1, 0, 0) \cdot \vec{v} \\ \vec{v} = (1, 0, 0) \cdot \vec{v} \end{cases} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \vec{v} = \vec{v} \\ & \begin{cases} (0, 1, 0) \cdot \vec{v} = (0, 1, 0) \cdot \vec{v} \\ (0, 1, 0) \cdot \vec{v} = (0, 1, 0) \cdot \vec{v} \\ \vec{v} = (0, 1, 0) \cdot \vec{v} \end{cases} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \vec{v} = \vec{v} \\ & \begin{cases} (0, 0, 1) \cdot \vec{v} = (0, 0, 1) \cdot \vec{v} \\ (0, 0, 1) \cdot \vec{v} = (0, 0, 1) \cdot \vec{v} \\ \vec{v} = (0, 0, 1) \cdot \vec{v} \end{cases} \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \cdot \vec{v} = \vec{v} \end{aligned}$$

$$0 = (1, 0, 0) \cdot \vec{v} \rightarrow \vec{v} = (1, 0, 0) \cdot \vec{v} \quad (1)$$

$$\begin{aligned} & (1, 0, 0) \cdot \vec{v} = (1, 0, 0) \cdot \vec{v} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \vec{v} = \vec{v} \\ & (0, 1, 0) \cdot \vec{v} = (0, 1, 0) \cdot \vec{v} \end{aligned}$$

$$(0, 1, 0) \cdot \vec{v} = (0, 1, 0) \cdot \vec{v}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} & (1, 0, 0) \cdot \vec{v} = (1, 0, 0) \cdot \vec{v} \\ & \vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) Given U_1, U_2 are independent $\rightarrow$ uniform RV on $(0, 1)$

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}; \quad x_1 = U_1, \quad x_2 = U_2, \quad x_3 = U_1 + U_2$$

$$\text{Cov}(U_1 + U_2, U_1) = \text{Cov}(U_1, U_1) = \frac{1}{12}$$

$$\text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) = \frac{1}{6}$$

$$\rho = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix} \leftarrow \begin{cases} \text{Corr}(U_1, U_2) = 0 \\ \text{Corr}(U_1, U_1) = \text{Corr}(U_2, U_2) = 1 \\ \text{Corr}(U_1, U_1 + U_2) = \text{Corr}(U_2, U_1 + U_2) \\ = \frac{1/\sqrt{2}}{\sqrt{1/12 \times 1/6}} = 1/\sqrt{2} \end{cases}$$

$|\rho| = 0$
 $\rightarrow$ LD columns

(b) Given $\langle U_1, U_2, U_1 + U_2 \rangle$ $\mathbf{X} \mathbf{v} = \lambda \mathbf{v}$ $(\mathbf{X} - \lambda \mathbf{I}) \mathbf{v} = \mathbf{0}$

$$\mathbf{X} - \lambda \mathbf{I} = \begin{bmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{bmatrix} = \begin{aligned} &(1-\lambda)^3 - 1/\sqrt{2} (1/\sqrt{2} (1-\lambda)) \\ &+ 1/\sqrt{2} (-1/\sqrt{2} (1-\lambda)) \end{aligned}$$

Largest eigen value $\lambda = 2$

$$\begin{bmatrix} v_1 + v_3/\sqrt{2} \\ v_2 + v_3/\sqrt{2} \\ (v_1 + v_2)/\sqrt{2} + v_3 \end{bmatrix} = \begin{bmatrix} 2v_1 \\ 2v_2 \\ 2v_3 \end{bmatrix}$$

$$\begin{aligned} &= (1-\lambda)^3 - (1-\lambda) \\ &= (1-\lambda) \left((1-\lambda)^2 - 1 \right) \\ &\quad \lambda = 1 \quad \lambda = 0, \lambda = 2 \end{aligned}$$

$$\rightarrow v_1 = v_3/\sqrt{2}; \quad v_2 = v_3/\sqrt{2} \quad \left\{ \begin{array}{l} v_1 = v_2 \\ \sqrt{2} v_1 = v_3 \end{array} \right.$$

$$\begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} \quad \text{--or--} \quad \begin{bmatrix} 1/2 \\ 1/2 \\ 1/\sqrt{2} \end{bmatrix}$$

✓ after normalization

$$\therefore \text{Var} \left(\frac{1}{2}U_1 + \frac{1}{2}U_2 + \frac{1}{\sqrt{2}}(U_1 + U_2) \right)$$

$$= \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right)^2 \text{Var}(U_1) + \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right)^2 \text{Var}(U_2)$$

$$= 1.4571 \times \frac{1}{6} = 0.2428$$

Var of first principle component

maximum possible
variance of
Linear combination b/w
 $U_1, U_2, U_1 + U_2$ with
standardized eigen vectors.

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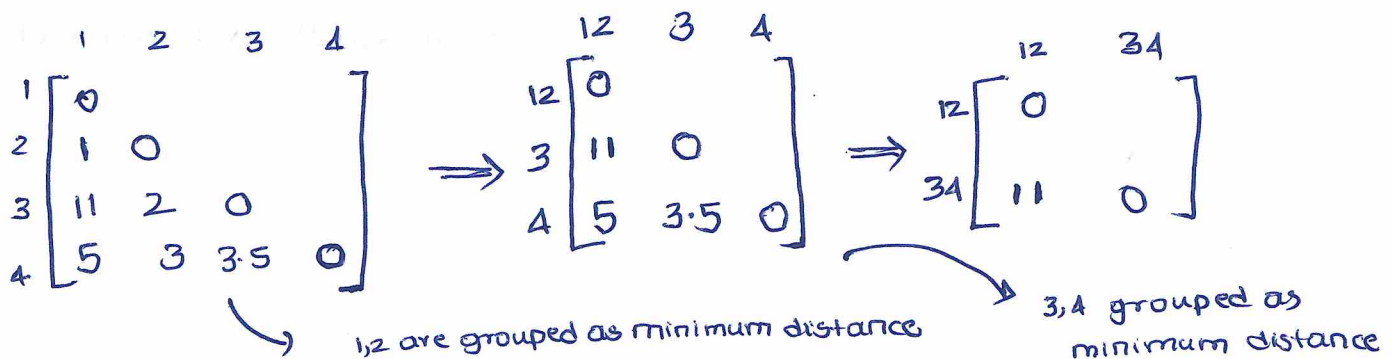
3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a)

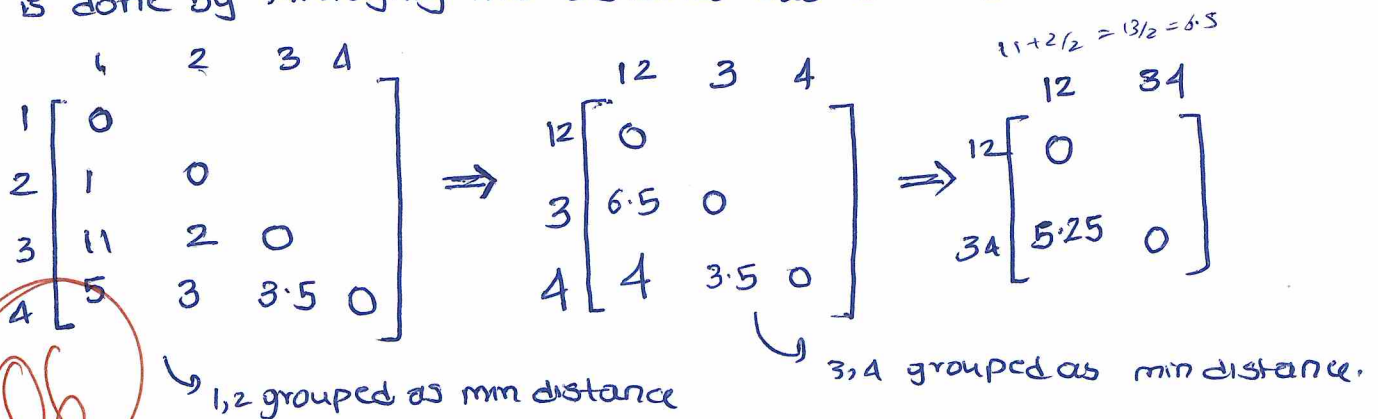
(i) Complete linkage hierarchical clustering
is done by comparing maximum distance b/w clusters.



Finally we group 3,4 to obtain the fully clustered 1234
with 12

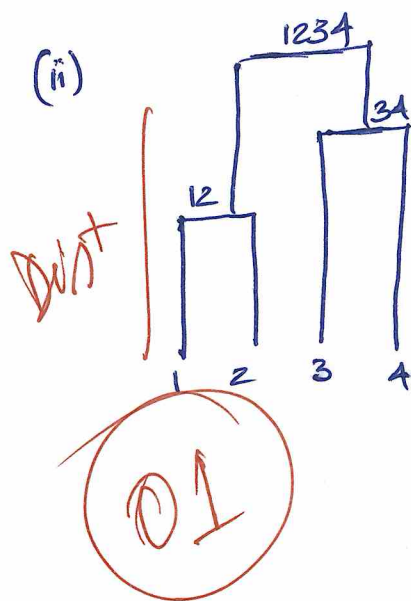
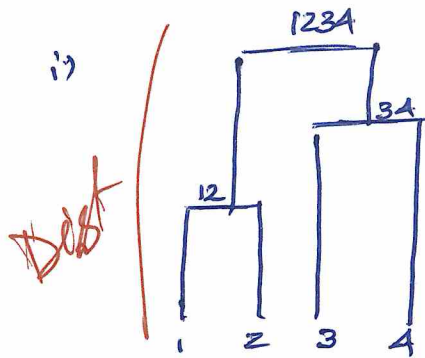
(ii) Average linkage hierarchical clustering

is done by Averaging the distance b/w clusters.



Finally we group 12, 34 to obtain fully clustered 1234

b)



The dendrograms turn out to be similar but the underlying values that represent inter-cluster distances remain different & it is possible that Complete linkage is more sensitive to outliers than Average linkage method.

ADDITIONAL PAGES FOR ANSWERS

